

Properties	Instruction			Training Datasets		
	Alpaca-GPT4	OpenOrca	Synth. Math-Instruct	Alignment		
Total # Samples	52K	2.91M	126K	Orca DPO Pairs	Ultrafeedback Cleaned	Synth. Math-Alignment
Maximum # Samples Used	52K	100K	52K	12.9K	60.8K	126K
Open Source	O	O	X	O	O	X

Table 1: Training datasets used for the instruction and alignment tuning stages, respectively. For the instruction tuning process, we utilized the Alpaca-GPT4 (Peng et al., 2023), OpenOrca (Mukherjee et al., 2023), and Synth. Math-Instruct datasets, while for the alignment tuning, we employed the Orca DPO Pairs (Intel, 2023), Ultrafeedback Cleaned (Cui et al., 2023; Ivison et al., 2023), and Synth. Math-Alignment datasets. The ‘Total # Samples’ indicates the total number of samples in the entire dataset. The ‘Maximum # Samples Used’ indicates the actual maximum number of samples that were used in training, which could be lower than the total number of samples in a given dataset. ‘Open Source’ indicates whether the dataset is open-sourced.

pretraining to quickly recover performance. We attribute the success of DUS to reducing such discrepancies in both the depthwise scaling and the continued pretraining steps. We also hypothesize that other methods of depthwise scaling could also work for DUS, as long as the discrepancy in the scaled model is sufficiently contained before the continued pretraining step.

Comparison to other up-scaling methods. Unlike Komatsuzaki et al. (2022), depthwise scaled models do not require additional modules like gating networks or dynamic expert selection. Consequently, scaled models in DUS do not necessitate a distinct training framework for optimal training efficiency, nor do they require specialized CUDA kernels for fast inference. A DUS model can seamlessly integrate into existing training and inference frameworks while maintaining high efficiency.

3 Training Details

After DUS, including continued pretraining, we perform fine-tuning of SOLAR 10.7B in two stages: 1) instruction tuning and 2) alignment tuning.

Instruction tuning. In the instruction tuning stage, the model is trained to follow instructions in a QA format (Zhang et al., 2023b). We mostly use open-source datasets but also synthesize a math QA dataset to enhance the model’s mathematical capabilities. A rundown of how we crafted the dataset is as follows. First, seed math data are collected from the Math (Hendrycks et al., 2021) dataset only, to avoid contamination with commonly used benchmark datasets such as GSM8K (Cobbe et al., 2021). Then, using a process similar to MetaMath (Yu et al., 2023), we rephrase the questions and answers of the seed math data. We use the resulting rephrased question-answer pairs as a QA dataset

and call it ‘Synth. Math-Instruct’.

Alignment tuning. In the alignment tuning stage, the instruction-tuned model is further fine-tuned to be more aligned with human or strong AI (e.g., GPT4 (OpenAI, 2023)) preferences using direct preference optimization (DPO) (Rafailov et al., 2023). Similar to the instruction tuning stage, we use mostly open-source datasets but also synthesize a math-focused alignment dataset utilizing the ‘Synth. Math-Instruct’ dataset mentioned in the instruction tuning stage.

The alignment data synthesis process is as follows. We take advantage of the fact that the rephrased question-answer pairs in Synth. Math-Instruct data are beneficial in enhancing the model’s mathematical capabilities (see Sec. 4.3.1). Thus, we speculate that the rephrased answer to the rephrased question is a better answer than the original answer, possibly due to the interim rephrasing step. Consequently, we set the rephrased question as the prompt and use the rephrased answer as the chosen response and the original answer as the rejected response and create the {prompt, chosen, rejected} DPO tuple. We aggregate the tuples from the rephrased question-answer pairs and call the resulting dataset ‘Synth. Math-Alignment’.

4 Results

4.1 Experimental Details

Training datasets. We present details regarding our training datasets for the instruction and alignment tuning stages in Tab. 1. We do not always use the entire dataset and instead subsample a set amount. Note that most of our training data is open-source, and the undisclosed datasets can be substituted for open-source alternatives such as the MetaMathQA (Yu et al., 2023) dataset.